FSD : Full-Stack Development Innovative Assignment



Report On : "Smart City Issue Reporting system"

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1. Executive Summary & Project Abstract

Municipal governance in modern developing cities faces a severe communication gap between tax-paying citizens and city administration. When potholes emerge, streetlights malfunction, pipelines burst, or garbage overflows, citizens have historically resorted to informal WhatsApp forwards, bureaucratic paper complaints, or non-responsive telephonic helplines. Complaints disappear into administrative black holes with zero status tracking, zero evidence verification, and zero accountability.

SmartCivic is an enterprise-grade, open-source municipal grievance management platform architected from first principles to bring radical transparency, strict SLA enforcement, and end-to-end accountability to urban service delivery. By fusing interactive Leaflet GIS geolocation, AI-based duplicate detection, strict 5-tier Role-Based Access Control (RBAC), mandatory photographic evidence verification, and an immutable 10-stage state machine, SmartCivic transforms civic governance into a responsive, measurable, and citizen-first operation.

> **Core Project Mission**
> To eliminate citizen disillusionment and administrative bottlenecks by guaranteeing evidence-backed grievance redressal under statutory SLA deadlines, verified by photographic proof and tracked live by citizens.

2. Real-World Civic Problems Addressed

• **The 'Black Hole' Syndrome (Lack of Visibility):** Citizens file grievances through generic portals or helplines but receive no tracking tokens, estimated resolution times, or assigned officer details. Over time, citizens lose faith in local governance and cease reporting urban hazards.

• **Duplicate Grievance Flooding & Back-Office Overwhelm:** When a prominent water pipe bursts or a major road crater forms, dozens of residents file separate complaints for the identical issue. Back-offices spend hundreds of man-hours manually weeding out duplicates rather than dispatching repair crews.

• **Ghost Redressals & Fraudulent Closures:** In conventional municipal setups, contractors or technicians routinely mark tickets as 'Completed' or 'Resolved' without executing any physical on-site repair. Because citizens cannot verify or challenge these closures, problems persist indefinitely.

• **Uniform Queue Delays for Critical Emergencies:** Traditional systems treat all grievances with a first-in, first-out (FIFO) queue. A minor park bench paint chip receives the same operational priority as an exposed live high-voltage electric wire or a contaminated water reservoir leak.

• **Departmental Silos & Finger-Pointing:** A collapsed storm drain affecting a roadway often leads to endless jurisdiction battles between Roads & Infrastructure and Water Supply & Sewerage. Without an automated routing engine, grievances bounce endlessly between departments.

• **Absence of Executive Geospatial Intelligence:** Municipal Commissioners and Ward Councilors lack spatial heatmaps to detect cluster patterns, infrastructure decay hotspots, or repeat contractor failures across city wards.

3. Engineered Solution & Innovation Highlights

✓ **GPS-Geotagged Evidence Intake:** Citizens pinpoint the exact geographic coordinate of the incident using an interactive OpenStreetMap/Leaflet picker with sub-meter precision, complemented by mandatory high-resolution photo evidence upload.

✓ **Proximity-Based AI Duplicate Detection:** Before a grievance is submitted, the system queries a 50-meter geospatial radius for active issues in the same category. If a duplicate exists, the citizen is prompted to upvote the existing ticket instead of creating redundant noise.

✓ **Strict 5-Tier RBAC Architecture:** Eliminates confusion by providing dedicated, tailored operational workspaces for Public Citizens, Department Triage Officers, Field Repair Technicians, Quality Supervisors, and Municipal Administrators.

✓ **Mandatory Photographic Before/After Proof:** Field technicians cannot close a work order without uploading geo-referenced photographic proof of the finished repair alongside physical materials deployed (e.g. cold-mix asphalt, LED fixtures, replacement valves).

✓ **Independent Quality Audit Desk:** Field Supervisors independently inspect technician proof before approving final resolution. If the work is substandard, the ticket is rejected back to the field crew for immediate rework.

✓ **Citizen Reopen Guarantee (48h Window):** Citizens retain final authority. Upon ticket closure, citizens can rate resolution satisfaction (1–5 stars) or reopen the ticket within 48 hours if the physical issue was not genuinely resolved.

✓ **Statutory SLA Countdown Matrix:** Every ticket is bound to a legally mandated resolution countdown (Critical: 6h, High: 24h, Medium: 48h, Low: 72h). Automated warnings trigger at 75–85% elapsed time, and breaches escalate directly to executive admin dashboards.

✓ **Immutable Forensic Audit Ledger:** Every state change, priority adjustment, worker assignment, and comment is permanently appended to a tamper-evident audit table capturing the actor, IP address, timestamp, and previous state.

4. Full Technical Architecture & Monorepo Structure

SmartCivic follows a decoupled client-server architecture designed for high scalability, rapid API response times, and clean separation of concerns. The backend exposes stateless RESTful APIs authenticated via JWT tokens, while the frontend is a Single Page Application (SPA) built with React 18, TypeScript, and Vite.

Layer	Technologies Deployed	Primary Architectural Responsibility
Presentation Layer	React 18, TypeScript, Vite, Tailwind CSS	Responsive UI, dynamic role sidebars, Leaflet GIS map rendering, real-time Recharts dashboards.
API & Routing Layer	Django REST Framework (DRF), SimpleJWT	CORS handling, JWT bearer authentication, route guards, serializer validation, pagination.
Business Logic Layer	Django 5.x Models & State Machine	10-stage lifecycle validation, proximity

		duplicate detection, SLA countdown calculation, audit logging.
Persistence Layer	SQLite / PostgreSQL, Django ORM	ACID transactions, indexed geospatial coordinates (lat/lng), foreign key integrity, media storage.
Geospatial GIS Engine	Leaflet 1.9, React-Leaflet, Leaflet.heat	Client-side GIS rendering, reverse geocoding fallback, ward heatmaps, cluster markers.
Media & Evidence	Python Pillow (PIL), Django Static/Media	Image optimization, format validation, before/after proof storage.

5. Complete Technology Stack Analysis

Component	Technology	Version	Rationale & Engineering Advantage
Backend	Python	3.12+	High-level expressive language with rich ecosystem for geospatial, data science, and web APIs.
Web Framework	Django	5.1.x	Enterprise-ready framework with built-in ORM, migration system, security middlewares, and admin panel.
API Framework	Django REST Framework	3.15.x	Industry standard for robust serializers, viewsets, token authentication, and status codes.
Auth Engine	django-rest-framework-simplejwt	5.3.x	Stateless, secure JSON Web Token authentication with access and refresh token rotation.
Frontend UI	React	18.3.x	Component-driven reactive UI architecture with

			Concurrent Mode and efficient virtual DOM reconciliation.
Language	TypeScript	5.6.x	Strict compile-time type safety preventing runtime null/undefined bugs across 150+ civic interfaces.
Build Tool	Vite	5.4.x	Next-generation ESM frontend bundler providing instant HMR and optimized production Rollup chunks.
CSS Styling	Tailwind CSS	3.4.x	Utility-first design token system enforcing strict municipal colors, responsive breakpoints, and animations.
GIS Maps	Leaflet & React-Leaflet	1.9.x	Lightweight, mobile-friendly interactive mapping engine with tile caching and coordinate reverse-lookup.
Data Viz	Recharts	2.12.x	Declarative SVG charting library for ward SLA compliance, department load, and incident trend curves.
Data Fetching	TanStack React Query	5.56.x	Server-state caching, automatic background invalidation, and seamless optimistic UI updates.

6. 5-Tier Role-Based Access Control (RBAC) Framework

SmartCivic implements a strict 5-tier operational hierarchy to ensure separation of municipal duties, preventing conflicts of interest and guaranteeing administrative checks and balances:

Role Name	Primary Persona	Dedicated Workspace Route	Permitted Operational Authority
CITIZEN	Public Citizen / Taxpayer	/citizen, /report, /track	Submit geotagged grievances, track real-time resolution timeline, upvote community issues, submit ratings, reopen tickets within 48h.
DEPARTMENT_OFFICER	Ward Department Officer	/officer	Review incoming grievances, accept or reject with cause, assign SLA priority, reroute between departments, dispatch work orders to field crews.
WORKER	Field Repair Technician	/worker	Receive digital job sheets, update dispatch status (Arrived, In Progress), log deployed repair materials, upload photographic proof of completion.
SUPERVISOR	Quality & Safety Auditor	/supervisor	Inspect on-site repair proof submitted by technicians, approve final closure, or reject back to field technician for immediate rework.
MUNICIPAL_ADMIN	Municipal Commissioner / Admin	/admin	Citywide grievance master registry, ward GIS heatmap analysis, executive departmental analytics, automated routing rules, SLA matrix tuning, forensic audit log.

7. 10-Stage Grievance Lifecycle State Machine & SLA Matrix

Every civic issue submitted to SmartCivic transitions through a deterministic, audit-logged 10-stage state machine. No stage may be bypassed, guaranteeing that every ticket possesses verifiable physical evidence before closure:

- **1. SUBMITTED:** Initial state upon citizen filing with GPS coordinates and initial damage photo.
- **2. UNDER_REVIEW:** Department Officer opens the grievance for preliminary verification and evidence triage.
- **3. ACCEPTED:** Officer formally validates the civic issue and classifies statutory SLA priority.
- **4. REJECTED:** Officer disallows invalid, out-of-jurisdiction, or fraudulent complaints with mandatory written remarks.
- **5. ASSIGNED:** Work order assigned to a specific field technician or repair crew with GPS routing.
- **6. ON_THE_WAY:** Technician accepts the dispatch and signals transit to the physical site.
- **7. IN_PROGRESS:** Technician arrives on-site and commences physical repair work.
- **8. DONE:** Repair completed. Technician uploads mandatory post-repair photo proof and logs materials.
- **9. VERIFIED:** Field Supervisor inspects proof and formally certifies quality standard fulfillment.
- **10. CLOSED / REOPENED:** Ticket archived upon citizen satisfaction, or reopened within 48h if physically unresolved.

Priority Class	Target Response	Statutory Deadline	Escalation Threshold	Typical Civic Scenarios
CRITICAL	Within 1 Hour	6 Hours	75% elapsed (4.5h)	Water pipeline burst, exposed live electrical cables, major sinkhole.
HIGH	Within 2 Hours	24 Hours	80% elapsed (19h)	Overflowing sewage, major arterial pothole, malfunctioning traffic signal.
MEDIUM	Within 4 Hours	48 Hours	80% elapsed (38h)	Non-functional streetlight, broken footpath slab, missed dumpster.
LOW	Within 8 Hours	72 Hours	85% elapsed (61h)	Faded lane markings, tree pruning requests, public park bench repair.

8. Detailed Module Walkthrough with System Screenshots

The following sections present high-resolution visual evidence of the running SmartCivic platform, showcasing each key subsystem, role dashboard, and GIS intelligence interface.

8.1 Public Citizen Landing Portal & Real-Time Metrics

The public landing page introduces citizens to SmartCivic's transparent mission. It features an interactive hero section, live municipal performance metrics (Total Grievances, Active Dispatches, Resolved Rate), quick-action buttons for complaint filing and ticket tracking, and an authentic civic image banner.

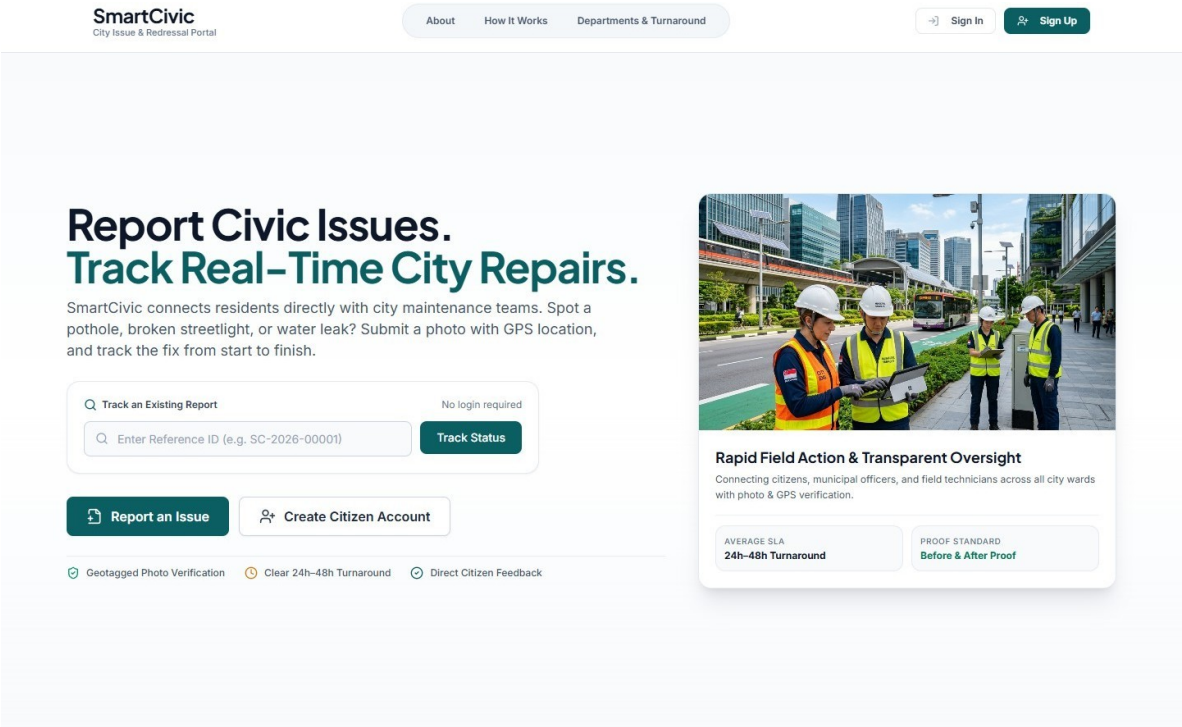


Figure: Public Landing Page Hero Section with Civic Service Indicators and Call-to-Actions

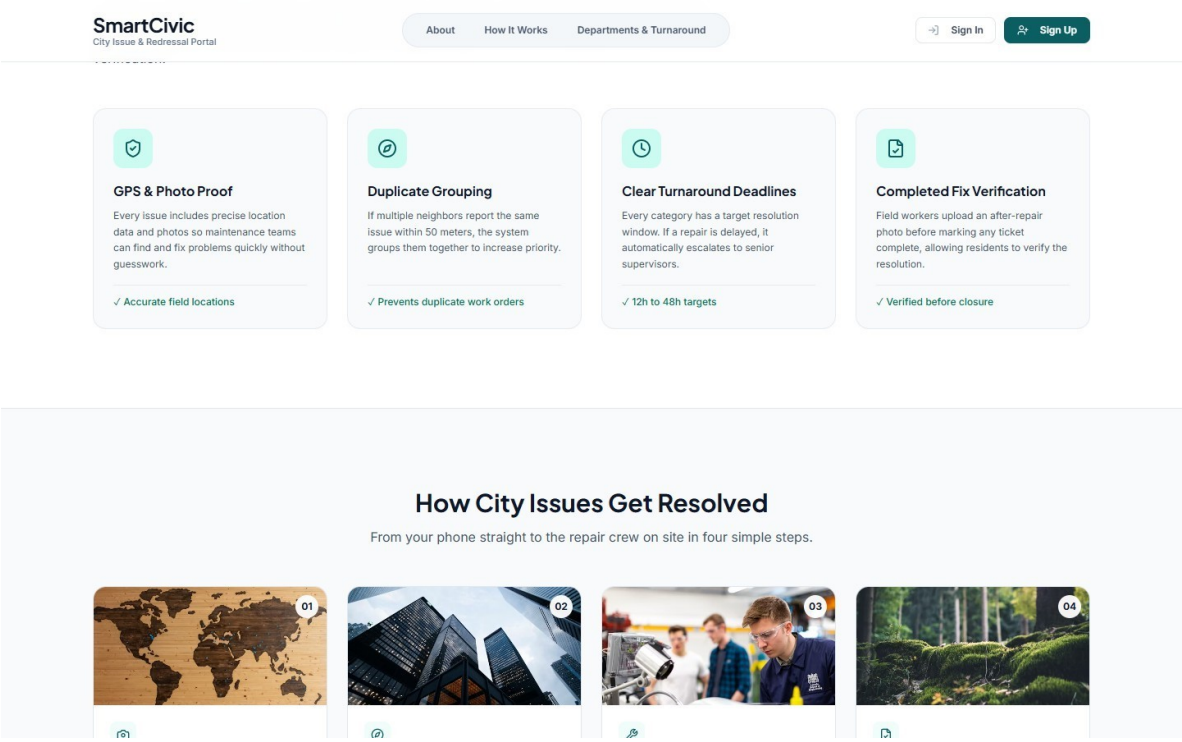


Figure: Civic Problem Category Directory with Real-Time Issue Volume Counts and Status Indicators

8.2 Citizen Complaint Filing with Leaflet GPS Geotagging

Filing a grievance is streamlined into an evidence-driven workflow. Citizens select the problem category (Pothole, Water Leak, Streetlight, etc.), click the interactive Leaflet map to pinpoint the exact GPS coordinates, type the incident description, and attach photographic evidence. The system automatically performs reverse geocoding to resolve the human-readable street address.

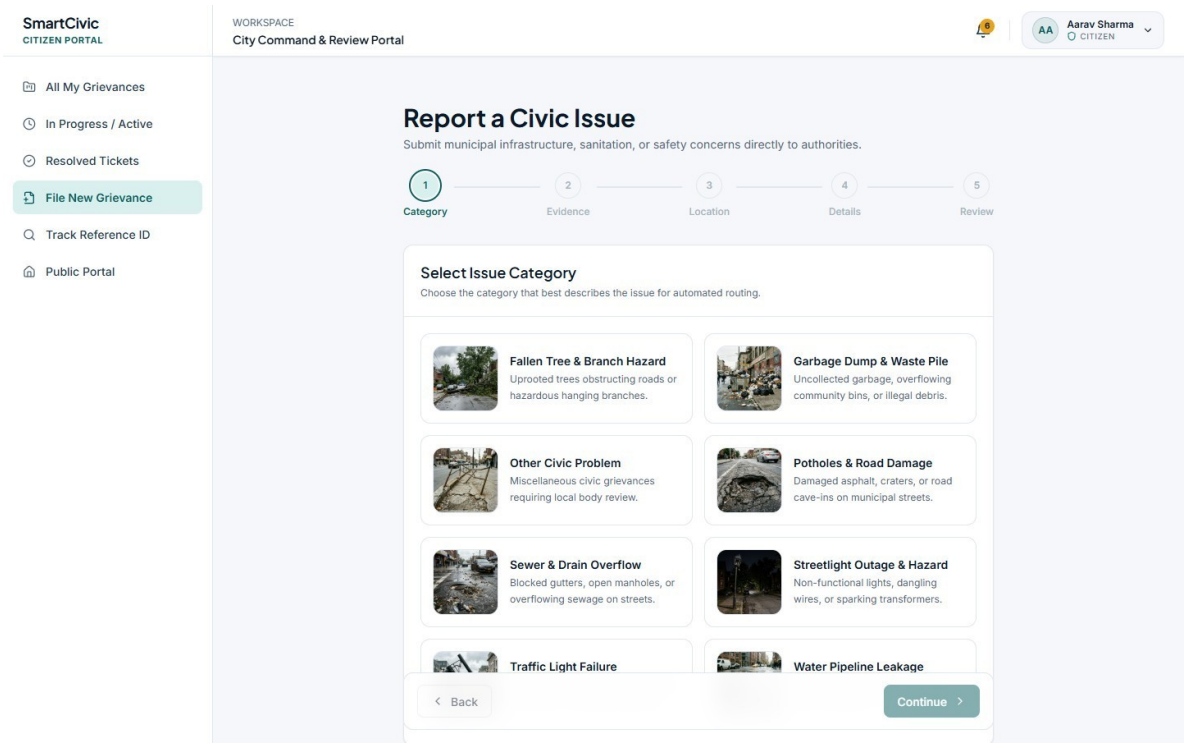


Figure: Citizen Grievance Submission Portal featuring Interactive Leaflet Map Geopicker & Photo Attachment

8.3 Transparent Public Grievance Tracking Timeline

Citizens can track any grievance without logging in simply by entering their unique reference token (e.g. SC-2026-00001). The tracking portal renders a real-time milestone timeline indicating who is handling the issue, current SLA status, assigned department, and live dispatch stage.

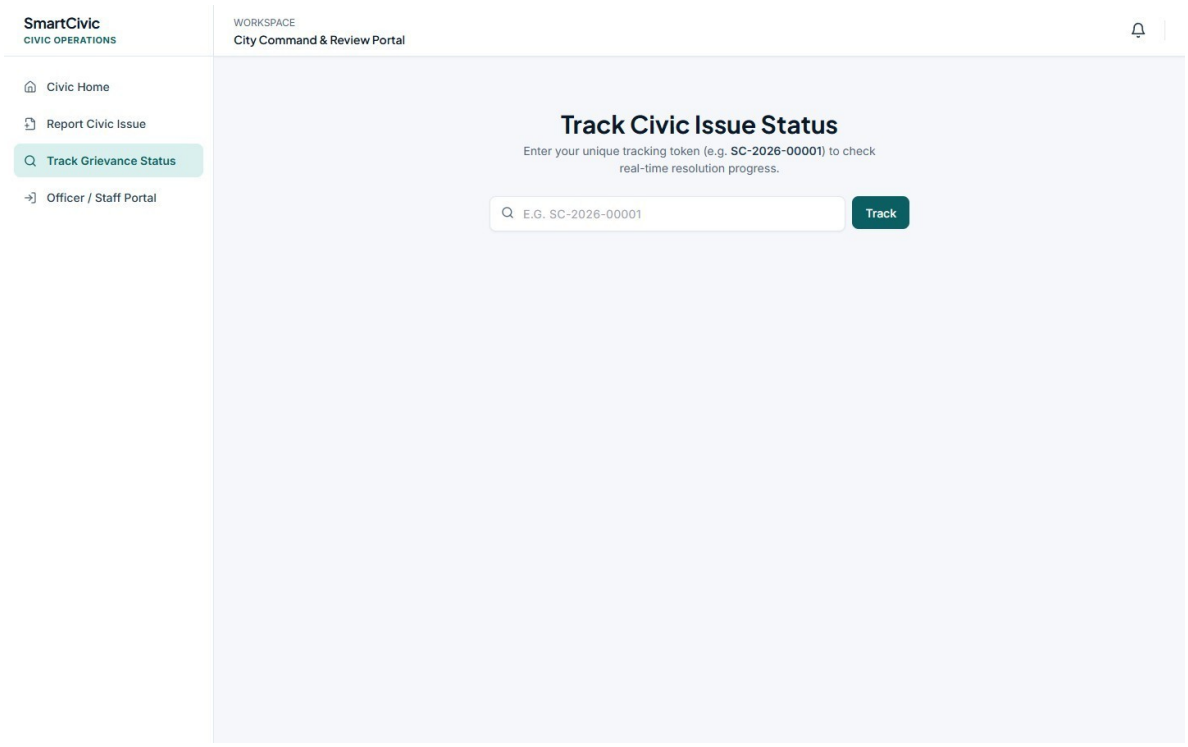


Figure: Public Tracking Lookup Portal with Human-Readable Token Input and Instant Search

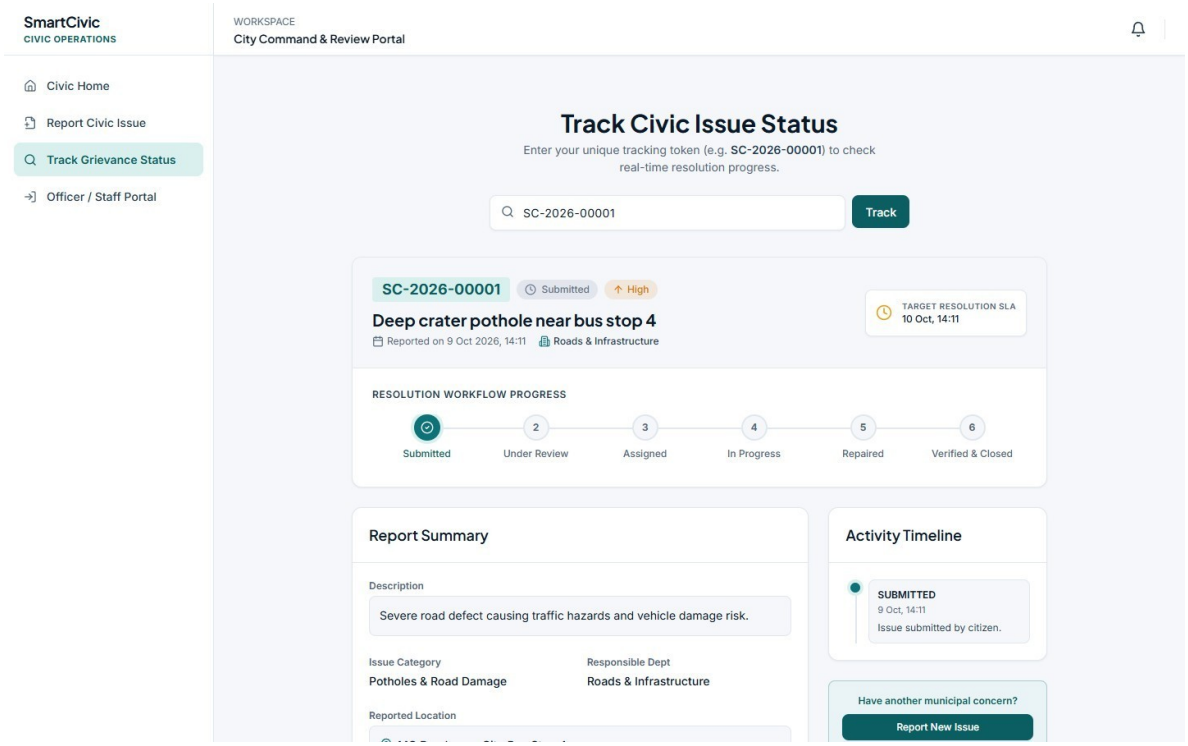


Figure: Real-Time Grievance Milestone Progress Timeline for Ticket SC-2026-00001

8.4 Multi-Role Authentication Desk & 1-Click Autofill

The authentication portal provides secure JWT login alongside an intuitive demo role selector. Evaluators and municipal staff can switch between Citizen, Officer, Supervisor, Worker, and Admin profiles with a single click, automatically populating department credentials and routing to the corresponding command desk.

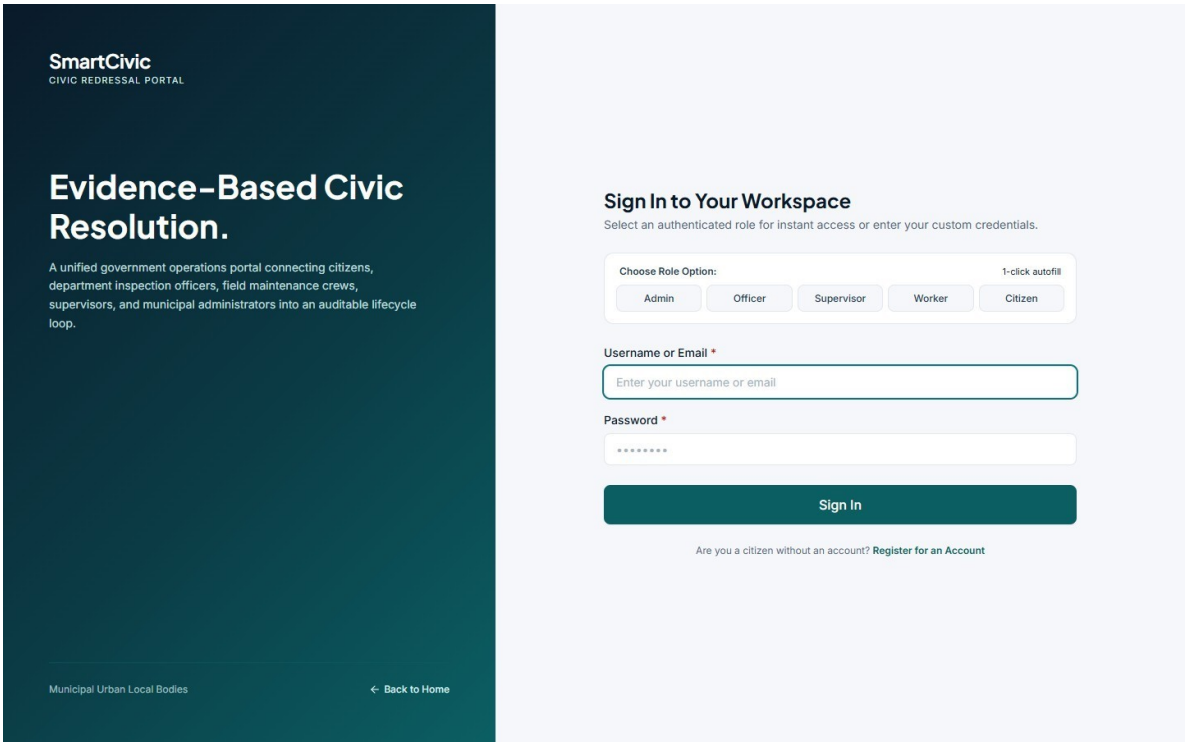


Figure: Multi-Role Authentication Portal with 1-Click Persona Autofill and Department Selector

8.5 Citizen Grievance Management Portal

Logged-in citizens access their personal grievance ledger, organized into tabbed views (All Grievances, Active, Resolved). Citizens can review historical tickets, inspect assigned technician progress, submit satisfaction ratings, or initiate a 48-hour reopen request if the physical issue was not resolved.

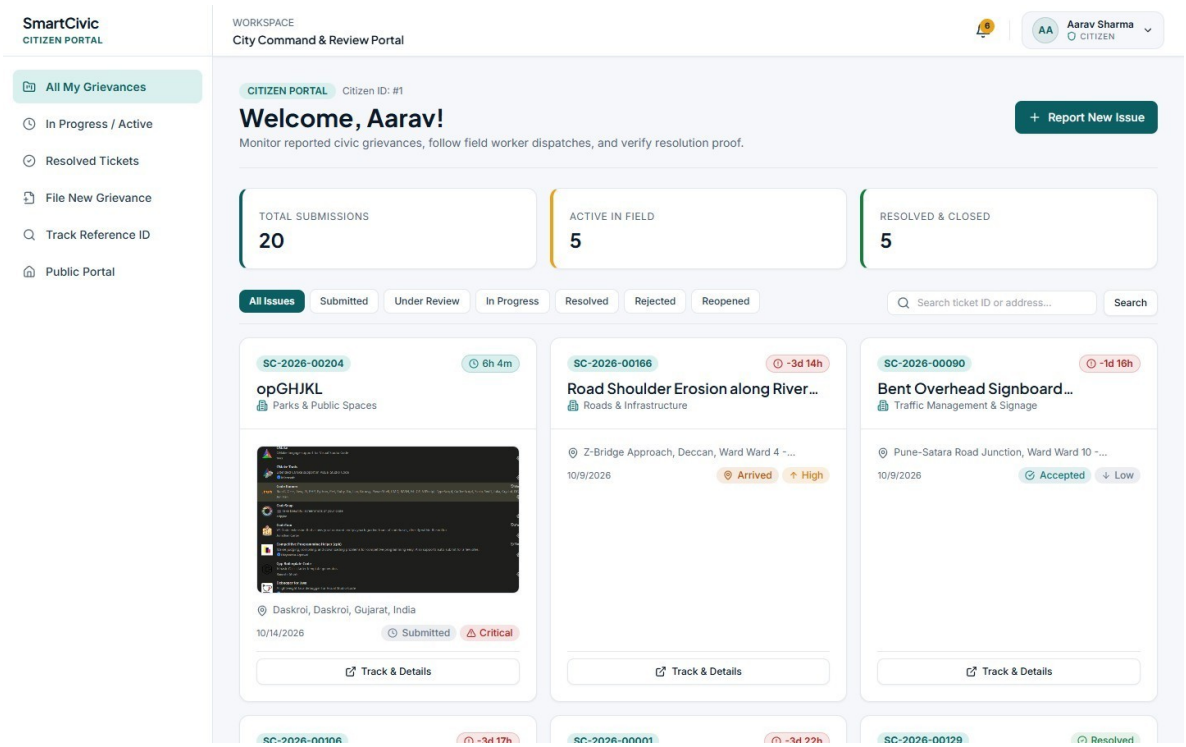


Figure: Citizen Operational Dashboard showing Personal Grievance Registry, Status Badges, and Quick Actions

8.6 Department Officer Command Desk & Intake Triage

Department Officers operate a dedicated triage queue for their specific civic domain (Roads, Water, Electrical, Sanitation). Officers review incoming photographic evidence, assign statutory SLA priorities, accept or reject issues, reroute misdirected tickets, and dispatch work orders to field crews with active SLA countdown timers.

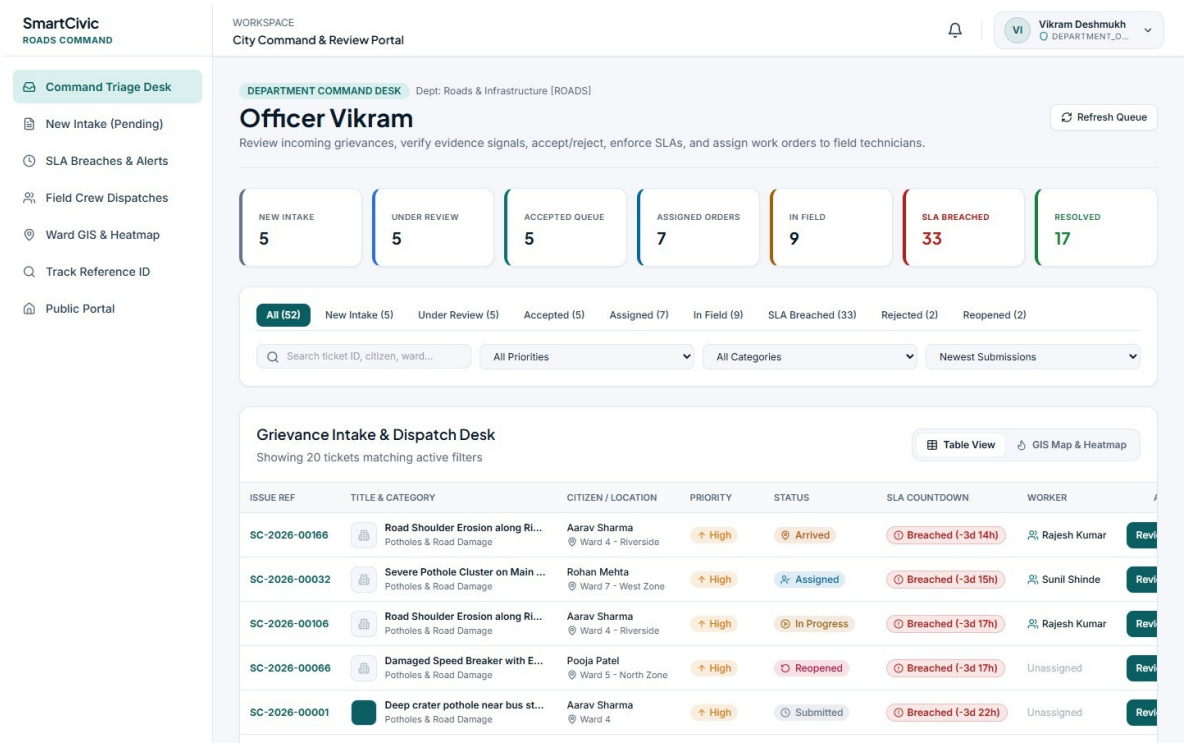


Figure: Department Officer Triage Command Desk showing Intake Queue, SLA Countdowns, and Work Orders

8.7 Department Ward GIS Heatmap & Geospatial Intelligence

Officers can toggle from the tabular queue into an interactive GIS Map view. Issues are rendered as color-coded pins and spatial density heatmaps across city wards, allowing officers to spot recurring infrastructure failure clusters.

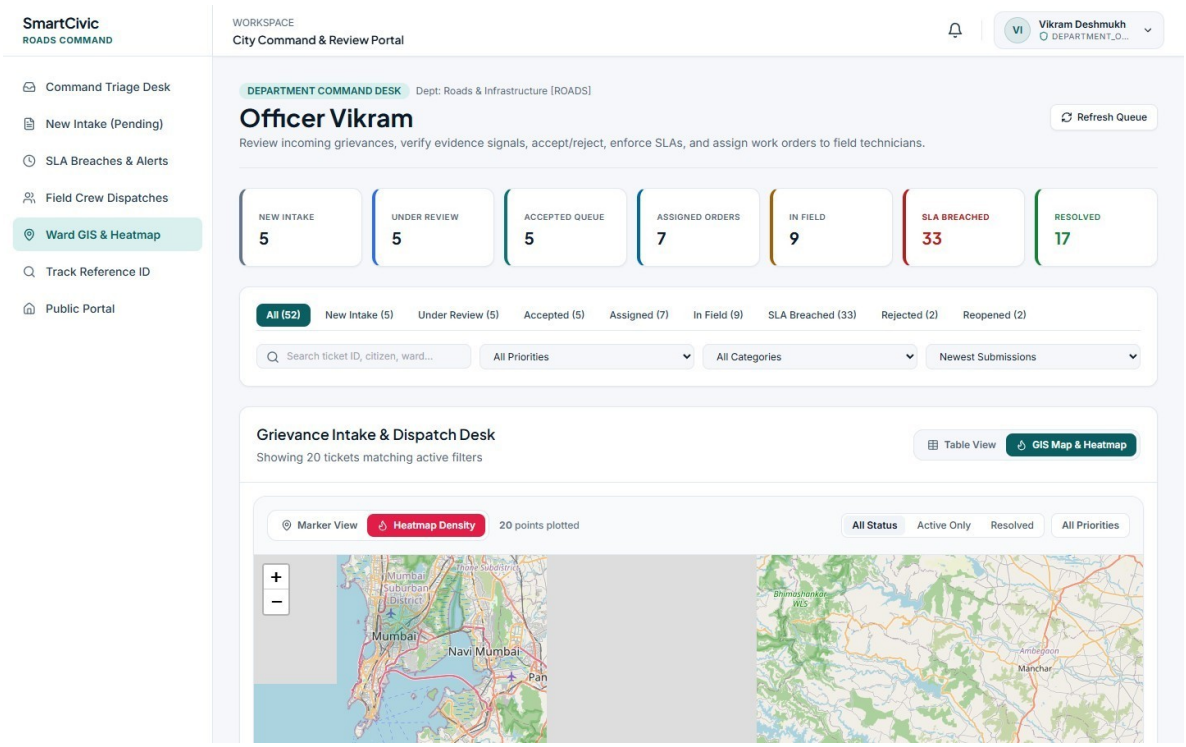


Figure: Department Officer Ward GIS Map with Interactive Issue Markers, Status Clustering, and Heatmap

8.8 Field Supervisor Quality Audit & Verification Desk

Field Supervisors provide independent quality assurance. When a technician finishes work, the ticket enters the Supervisor's desk. The supervisor audits the technician's post-repair photos, verifies deployed materials, and either certifies resolution or rejects the ticket back to the technician for immediate rework.

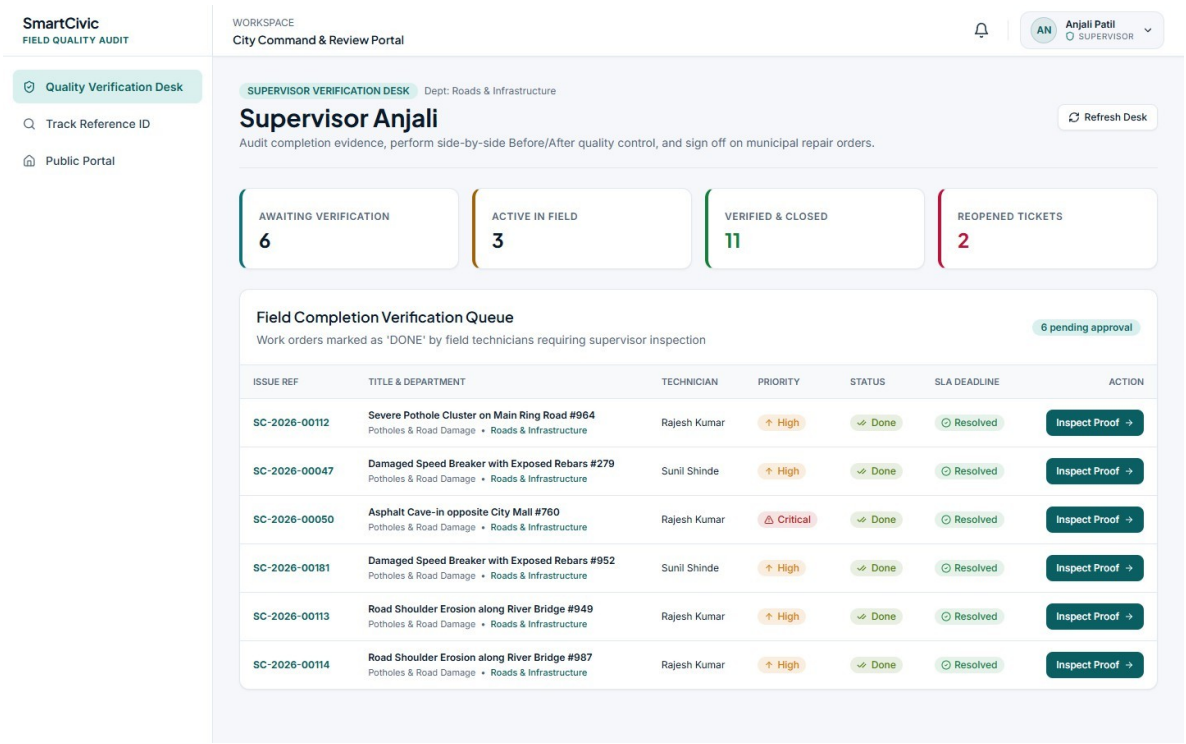


Figure: Field Supervisor Quality Audit Desk displaying Pending Approvals, Materials Deployed, and Evidence Inspection

8.9 Field Technician Digital Work Order & Proof Upload

Field Technicians use a mobile-optimized digital job sheet. Technicians update dispatch milestones ('On The Way' -> 'In Progress'), record repair notes, log deployed materials, and upload compulsory photographic evidence before a ticket can be completed.

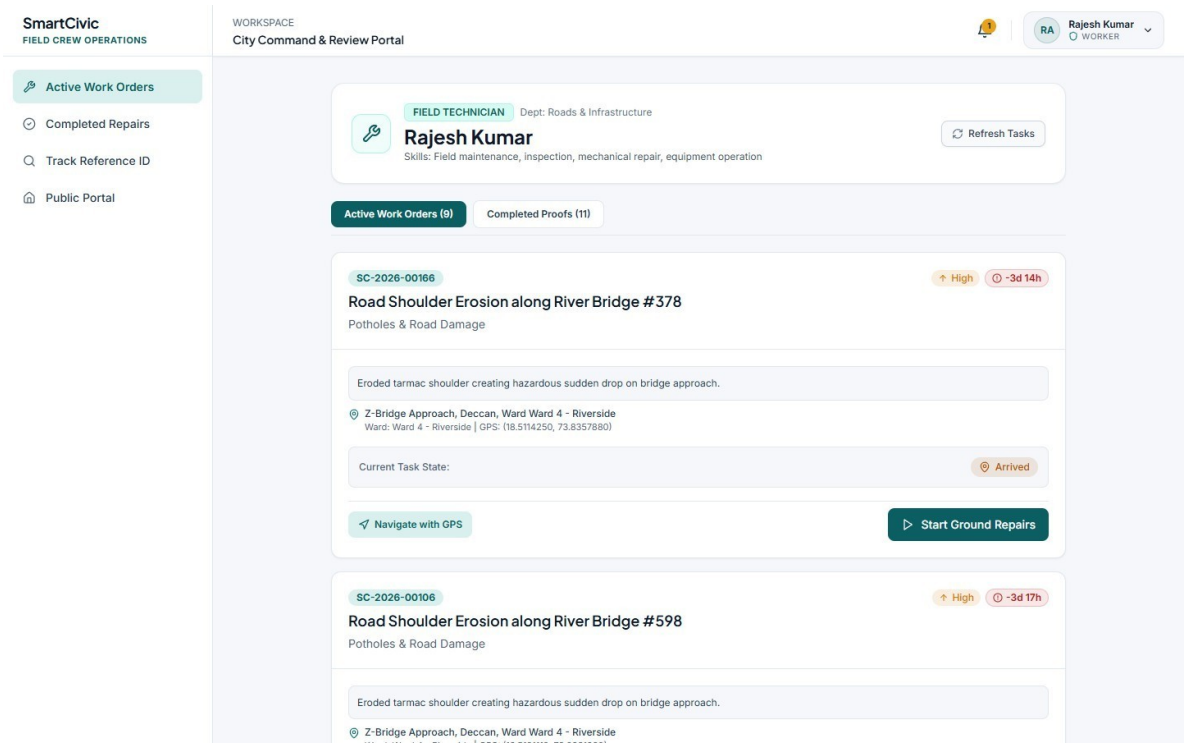


Figure: Field Technician Work Order Dashboard with Step-by-Step Dispatch Lifecycle and Evidence Upload Form

8.10 Municipal Admin Grievance Master & Citywide Hotspots

The Municipal Commissioner and City Administrators have macro-level oversight over all municipal departments. The Grievance Master view aggregates all city tickets with multi-criteria filtering across departments, wards, priorities, and SLA statuses.

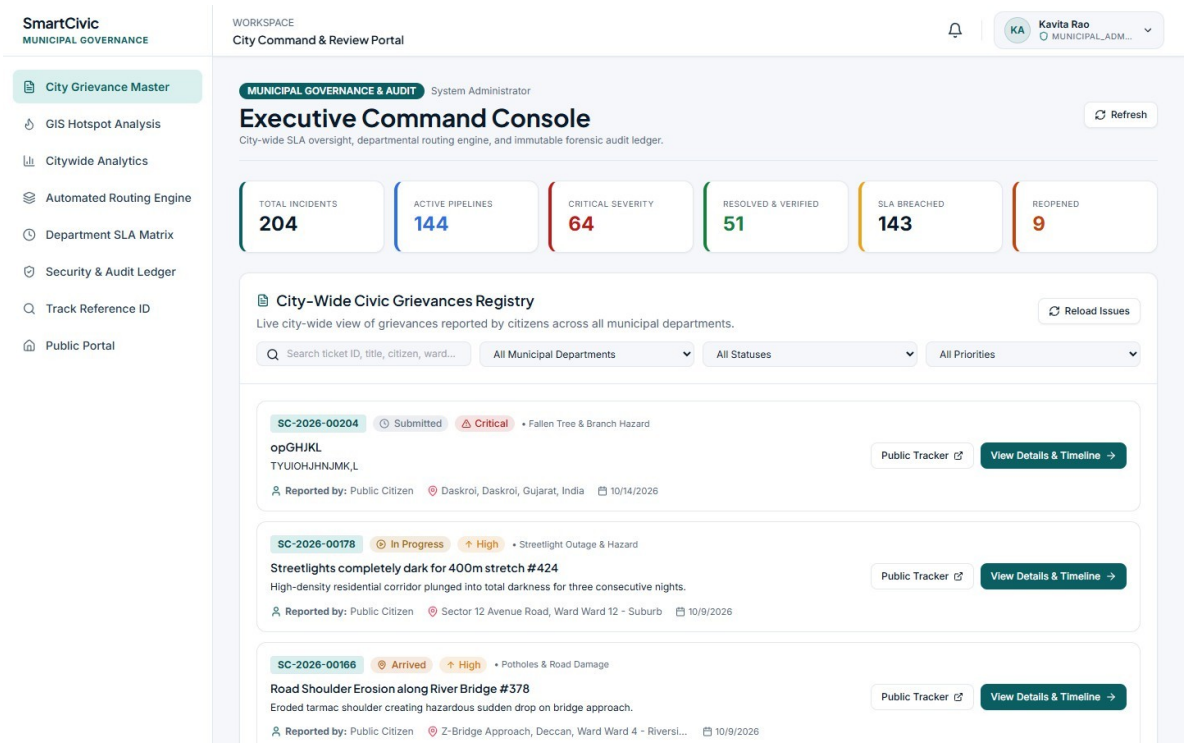


Figure: Municipal Admin Master Grievance Registry with Cross-Departmental Triage and Global Filters

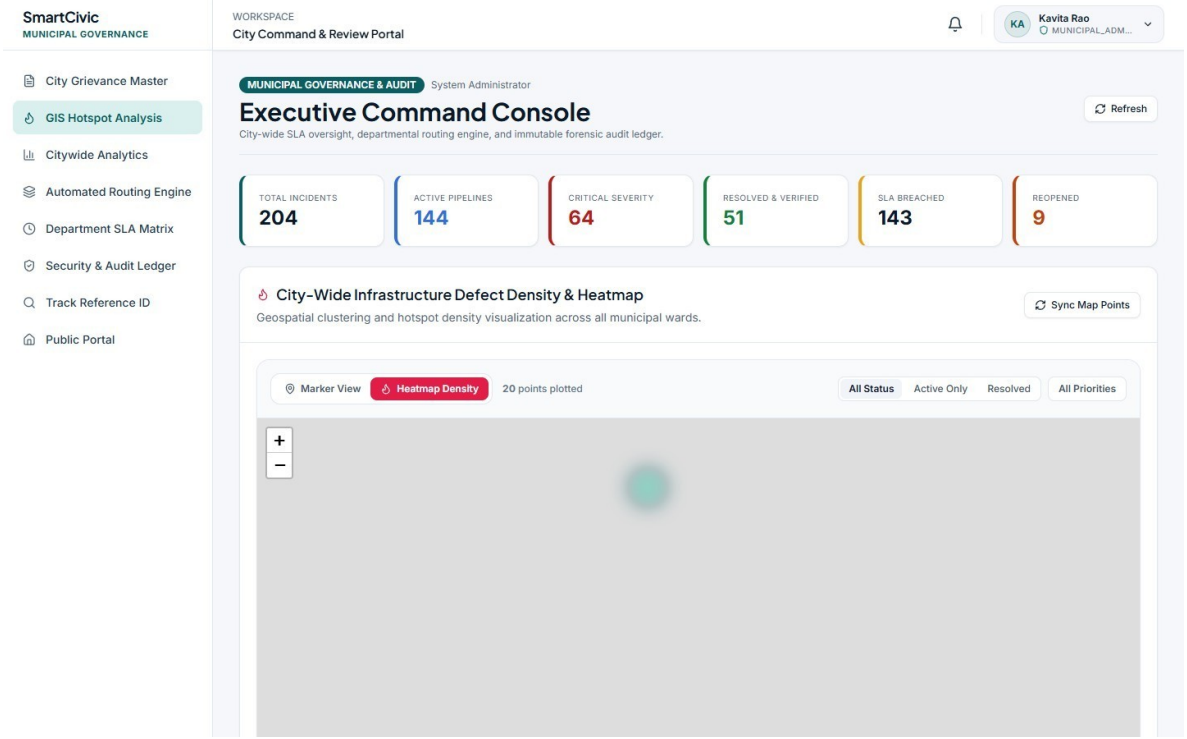


Figure: Citywide Hotspot Analysis GIS Interface displaying Infrastructure Stress Density Heatmaps

8.11 Executive Citywide Analytics & Department Workload

The Analytics dashboard provides executive business intelligence via interactive Recharts data visualizations: Ward-wise SLA compliance rates, department workload distribution, average resolution velocity, and monthly grievance trend curves.

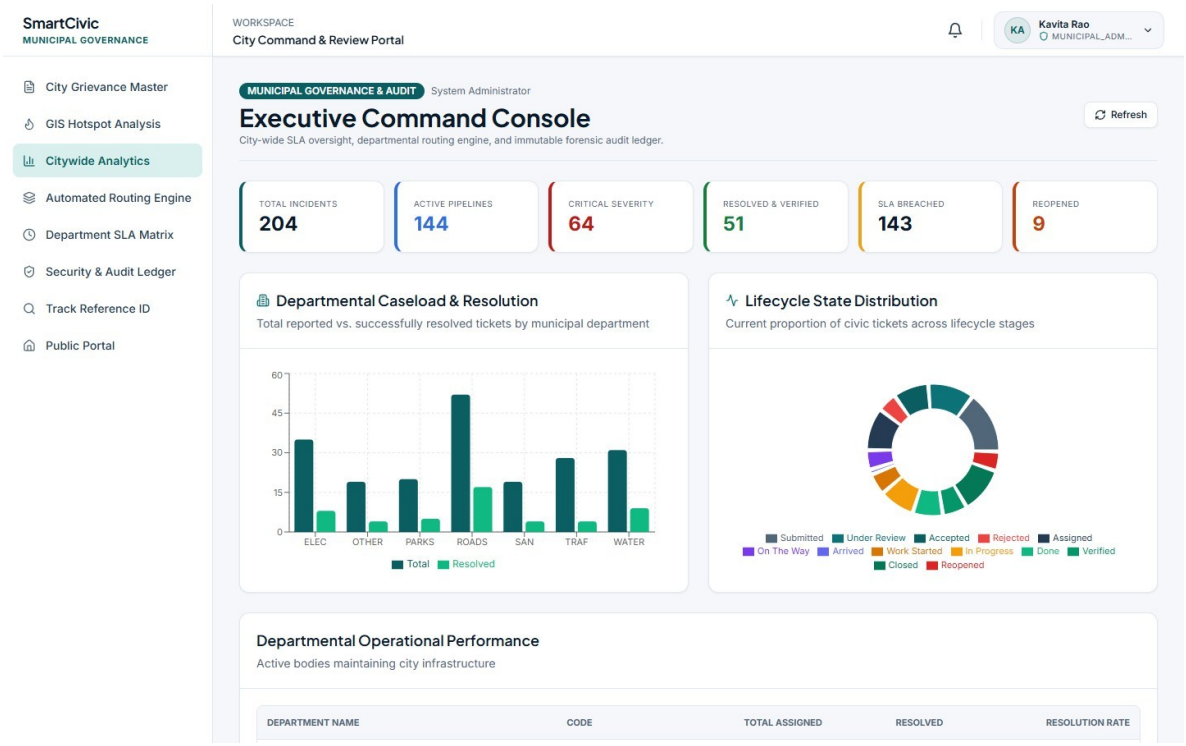


Figure: Executive Analytics Suite featuring Recharts Visualizations for SLA Compliance and Department Load

8.12 Immutable Forensic Security & Audit Ledger

SmartCivic maintains a zero-trust forensic audit log. Every lifecycle transition, priority alteration, worker assignment, and administrative note is permanently recorded with actor identity, timestamp, IP address, and state delta.

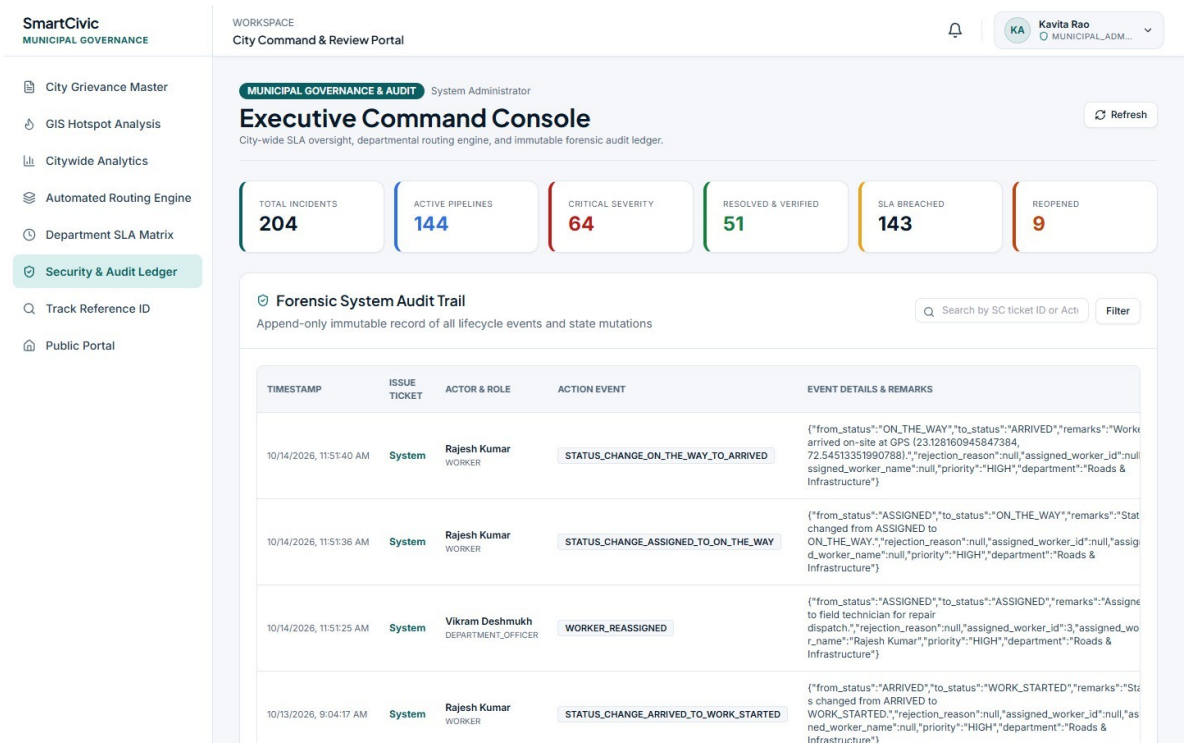


Figure: Immutable Forensic Security & Audit Ledger capturing Chronological Event History and Actor Footprints

9. Database Relational Schema & REST API Specifications

The backend database is structured to enforce referential integrity, geospatial indexing, and complete auditability:

Entity / Table	Key Attributes & Types	Relational Integrity & Constraints
User (accounts_user)	id, username, email, role, department_id, phone, is_verified	Custom user model with RBAC role enum and department foreign key.
Department (departments_department)	id, name, code, contact_email, head_officer_id	Municipal department definition (ROADS, WATER, ELEC, SAN, TRAFFIC, PARKS).
IssueCategory (issues_issuecategory)	id, name, code, default_priority, default_department_id	Civic issue taxonomy mapping each category to an SLA priority and routing target.
Issue (issues_issue)	issue_id, title, description, category_id, department_id,	Central grievance entity with geospatial lat/lng indices and lifecycle

	citizen_id, assigned_worker_id, latitude, longitude, address, status, priority, sla_deadline, is_breached	status enum.
Evidence (evidence_evidence)	id, issue_id, uploaded_by_id, file_path, evidence_type, created_at	Polymorphic evidence storage handling INITIAL_REPORT and RESOLUTION_PROOF images.
AuditTrail (audit_audittrail)	id, issue_id, actor_id, action, previous_status, new_status, remarks, ip_address, timestamp	Append-only forensic log capturing all lifecycle transitions.
Notification (notifications_notification)	id, recipient_id, issue_id, message, is_read, created_at	In-app alert delivery system notifying citizens and staff of state changes.

Key REST API Endpoint Catalog:

Method	Endpoint Path	Access Role	Operational Purpose
POST	/api/auth/token/	Public	Obtain JWT access & refresh token pair.
POST	/api/auth/register/	Public	Self-register new citizen or municipal staff.
GET	/api/auth/me/	Authenticated	Retrieve authenticated profile, role, and permissions.
GET	/api/issues/	Role-Filtered	Retrieve issues list (filtered by role scope and ward).
POST	/api/issues/	Citizen	Submit new geotagged grievance with photo.
GET	/api/issues/{id}/	Public / Auth	Retrieve issue details, SLA countdown, and evidence.
PATCH	/api/issues/{id}/triage/	Officer / Admin	Update priority, department routing, accept/reject.

PATCH	/api/issues/{id}/dispatch/	Officer / Sup	Assign work order to specific field technician.
PATCH	/api/issues/{id}/status/	Worker / Sup	Transition lifecycle stage (On the way -> Done -> Verified).
POST	/api/issues/{id}/evidence/	Worker	Upload post-repair photographic proof.
POST	/api/issues/{id}/reopen/	Citizen	Reopen ticket within 48h if physically unresolved.
GET	/api/audit/?issue_id={id}	Staff / Admin	Fetch immutable chronological event ledger for a ticket.

Persona Role	Username	Password	Department / Jurisdiction Scope
Municipal Admin	admin	pass123	Central Municipal Governance (Citywide oversight)
Department Officer	officer	pass123	Roads & Infrastructure Department
Field Supervisor	supervisor	pass123	Roads & Infrastructure Quality Verification
Field Technician	worker	pass123	Roads & Infrastructure Field Crew
Water Officer	officer_water	pass123	Water Supply & Sewerage Department
Electrical Officer	officer_elec	pass123	Electrical & Streetlighting Department
Public Citizen	citizen	pass123	Public Citizen / Taxpayer Account

Conclusion & Future Roadmap

SmartCivic represents a paradigm shift in open-source municipal software engineering. By replacing opaque paper trails and informal helplines with a deterministic, evidence-enforced digital workflow, the platform delivers verifiable accountability to urban administration. The integration of spatial GIS heatmaps, strict statutory SLAs, and photographic proof verification ensures that public taxes translate into tangible, high-quality urban infrastructure improvements.

Future Architectural Enhancements:

- **Automated Computer Vision Damage Assessment:** Integrating lightweight Edge AI models to classify pothole depth and pavement severity directly from citizen camera feeds.
- **Omnichannel Citizen Access (WhatsApp / Telegram Bots):** Enabling citizens without smartphones or internet access to report issues via interactive automated chat bots.
- **Automated Municipal Fleet Routing:** Optimizing repair truck routes based on geospatial cluster density to minimize municipal fuel expenditures.
- **Public Ward Leaderboard & Social Accountability:** Publishing monthly ward-by-ward SLA resolution rankings to foster healthy inter-departmental competition.